

LESSON 9

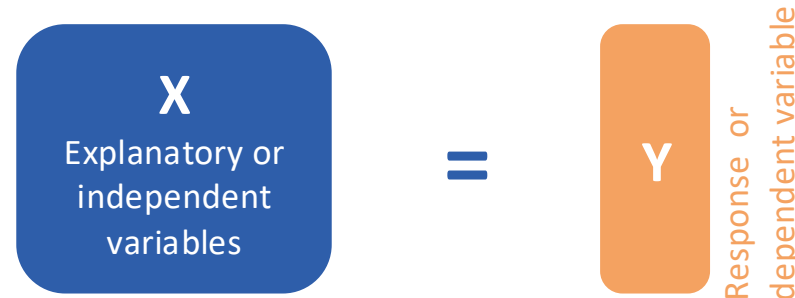
Regression models

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Definition

Regression models enable us to analyse the possible relationship between:

- The pattern of variation in a random variable (Y)
- The values of one or more variables on which the first variable depends, or may depend. (X)



This relationship is studied by establishing a model **whose parameters capture and quantify the effects under investigation**. These parameters are estimated from the available data

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Definition

The aim is generally to study the relationships between the distribution of Y and the values of the X_i .

We have:

- J observations of a random variable Y .
- The corresponding values of the variables $X_1, X_2, \dots, X_I$, on which the first depends or may depend.

$$\begin{matrix} x_{11}, \dots, x_{i1}, \dots, x_{I1} \\ \dots \\ x_{1j}, \dots, x_{ij}, \dots, x_{Ij} \\ \dots \\ x_{1J}, \dots, x_{iJ}, \dots, x_{IJ} \end{matrix} = \begin{matrix} y_1 \\ \dots \\ y_j \\ \dots \\ y_J \end{matrix}$$

Definition

Classical regression models assume that:

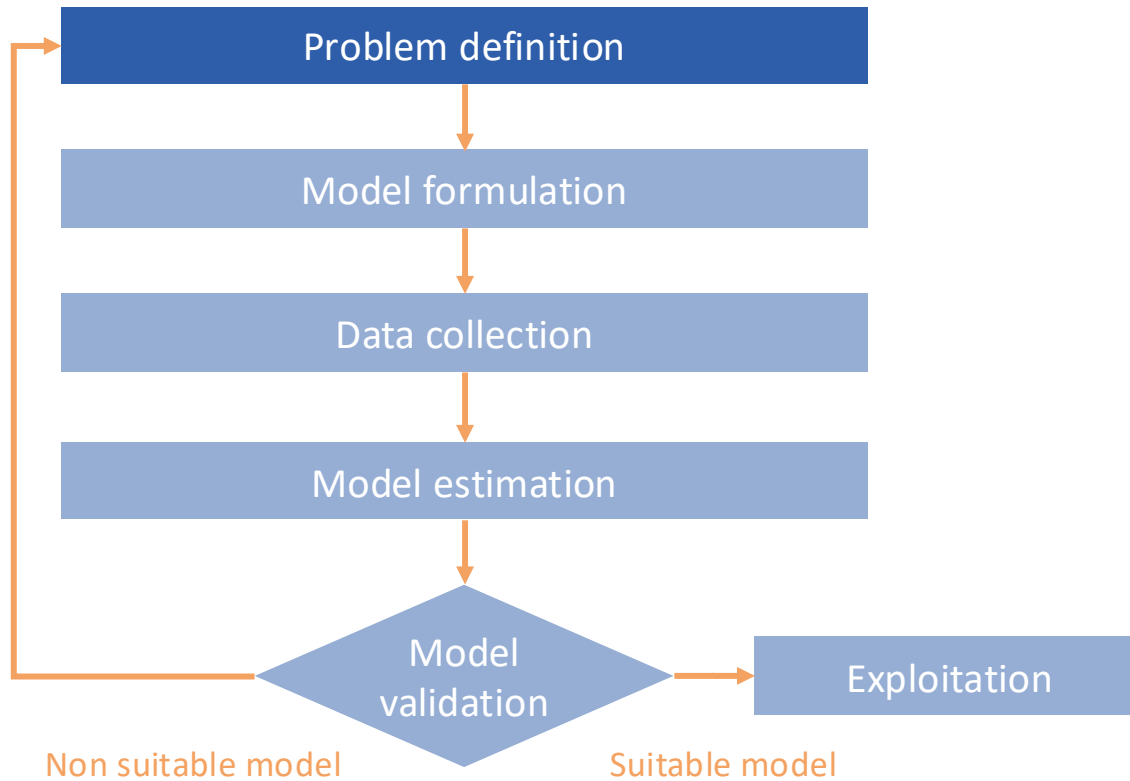
- Each observation y_j is the observed value of a normally distributed random variable Y :
- Constant and unknown variance $\sigma^2(Y)$
- The mean is a function of the observed values of X_i

$$Y \sim N(E(Y) = f(X_{1j}, \dots, X_{Ij}); \sigma^2(Y) = cte)$$

The possible effect of X_i on the distribution of Y is manifested in a change to the mean value of that dependent variable. The analysis can be generalised by also considering possible effects of X_i on the variance of Y .

$$\sigma^2(Y) = g(X_{1j}, \dots, X_{Ij})$$

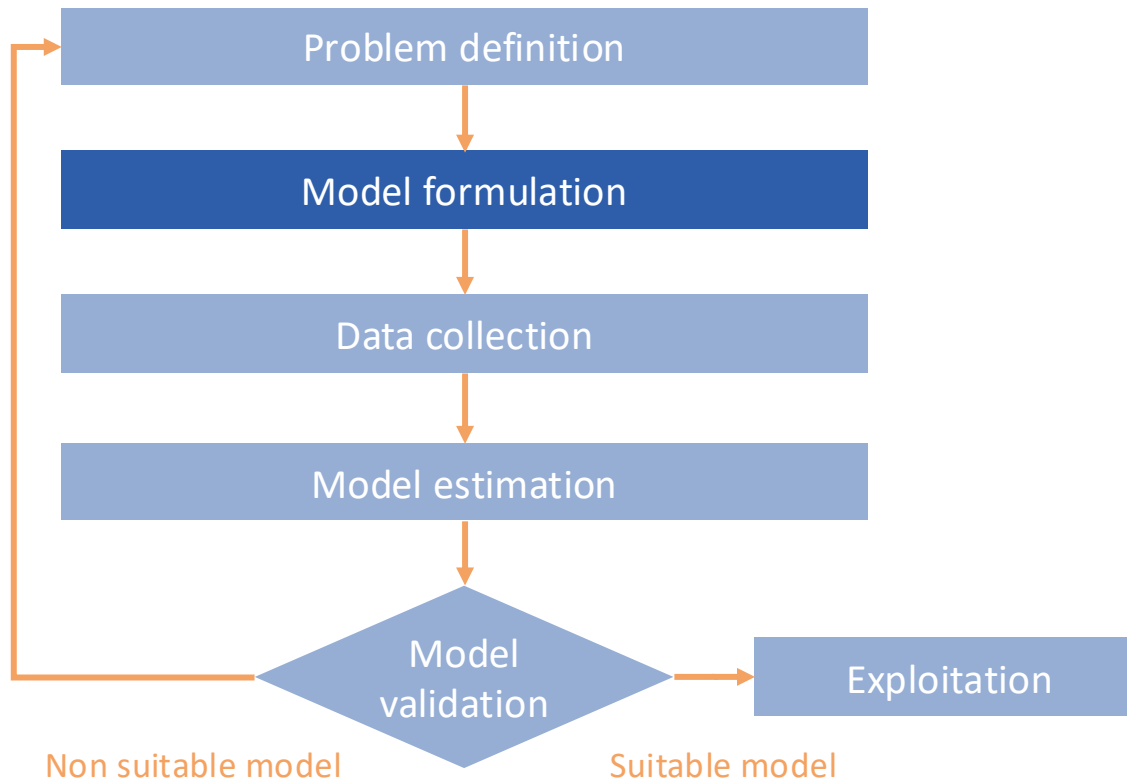
Stages of the study



The most delicate and crucial stage in the study of any real-world problem.

The most common mistake is to correctly solve the wrong problem (**Type III error**).

Stages of the study

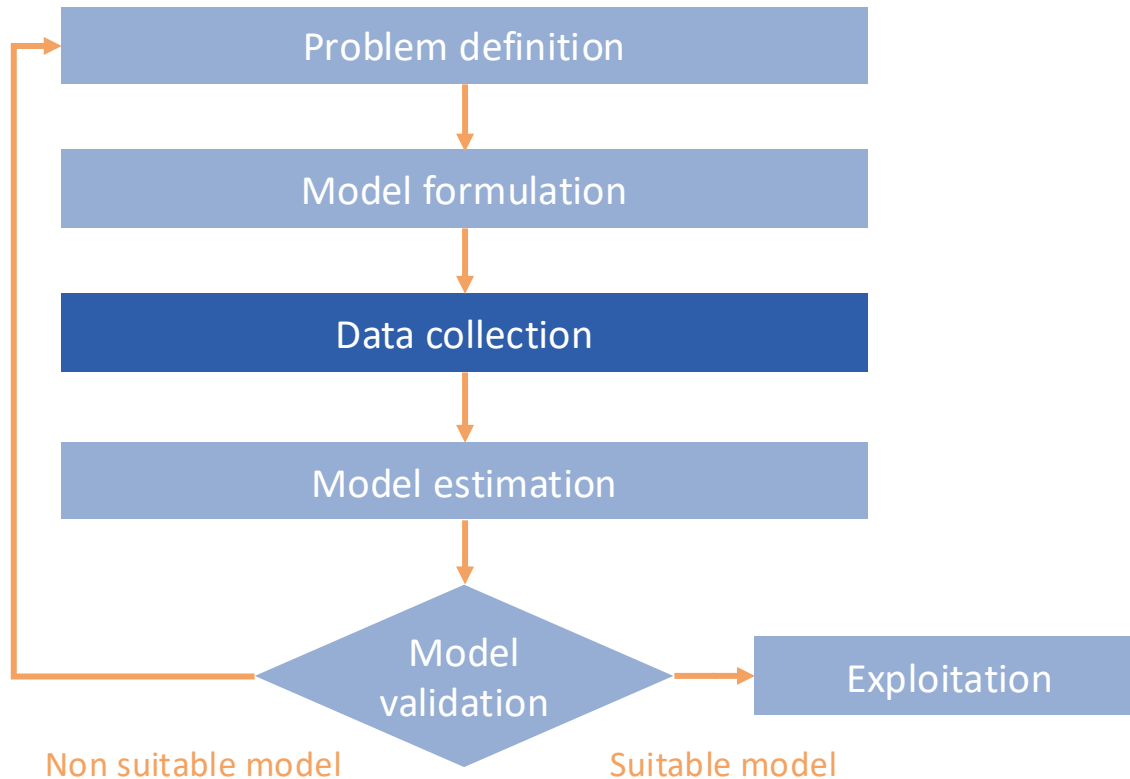


Definition of:

- The response variable (Y).
- The selection of explanatory (or independent) variables (X_i).
- The choice of the **analytical form of the model equation**.
- Whether or not **to include interactions**.
- Application (if necessary) of **transformations** to the variables.
- Interpretation of the **meaning of the parameters**.

By the end of this phase, it should be clear “what” each of the model’s parameters is, even if “the value” is not yet known.

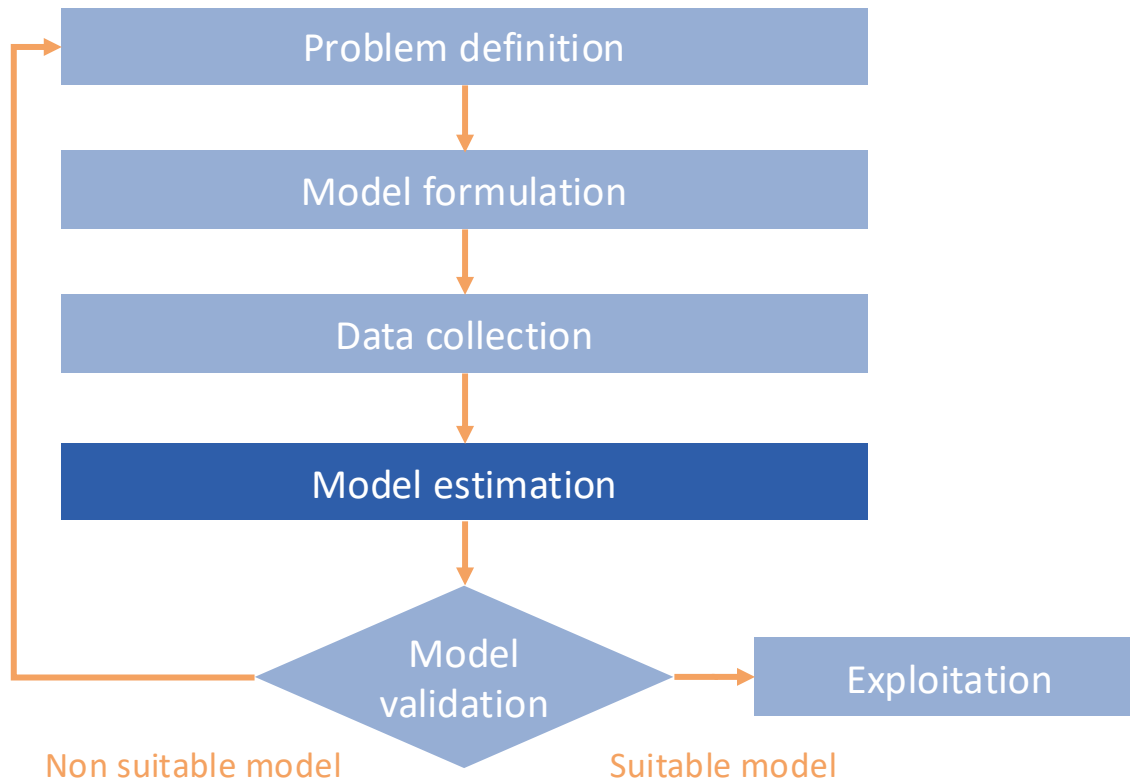
Stages of the study



Various techniques can be used:

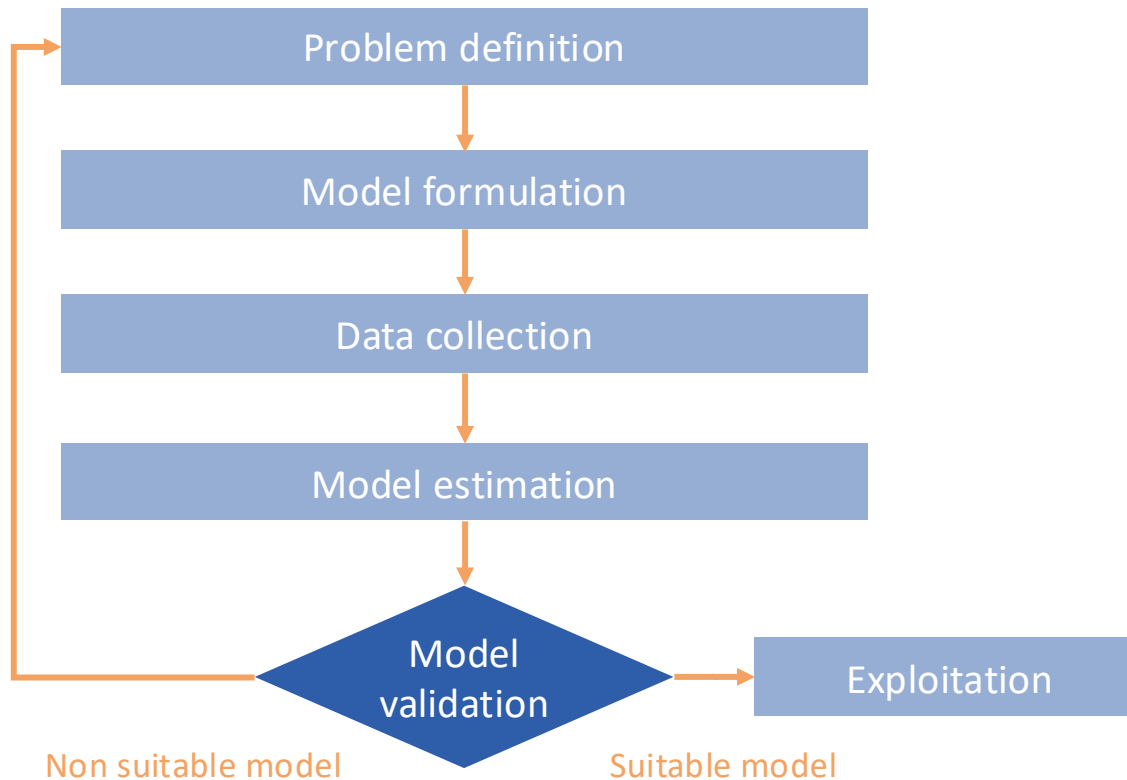
1. **Compilation of existing data.**
2. **Sampling:** used in observational sciences.
3. **Experimental design:** used in experimental sciences; it requires a clear understanding of the model to be estimated. Ideally, one should be able to control the explanatory variables.

Stages of the study



The information contained in the data is used to obtain the **most plausible values** for the model **parameters** (β_i coefficients of the regression equation and residual $\sigma_{residual}^2$), as well as to assess the **accuracy** of the estimates obtained.

Stages of the study

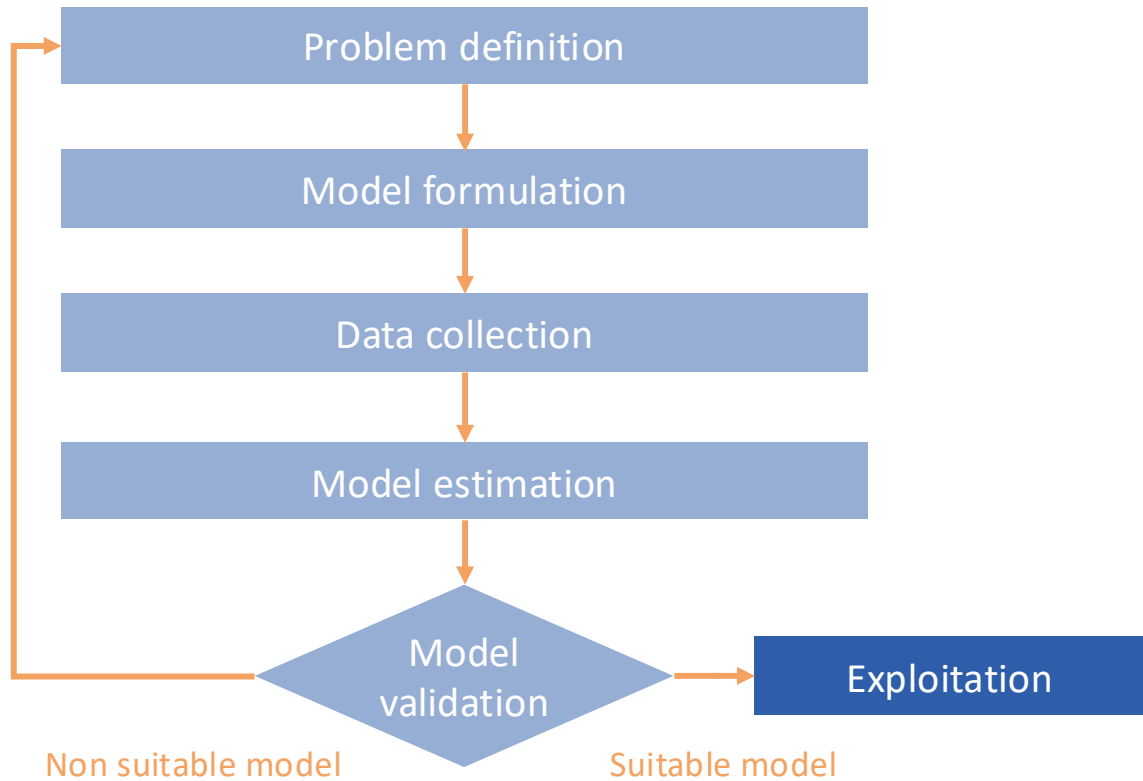


The aim is to make use of the information contained in the data not only to estimate the parameters of the proposed model, but also to question the validity of the assumptions on which that model is based. The technique we will examine is **residual analysis**.

- Are the data normally distributed?
- Are there any outliers that could significantly affect the study's conclusions?
- Is the chosen analytical method appropriate?

All models are false but... some models are useful.

Stages of the study



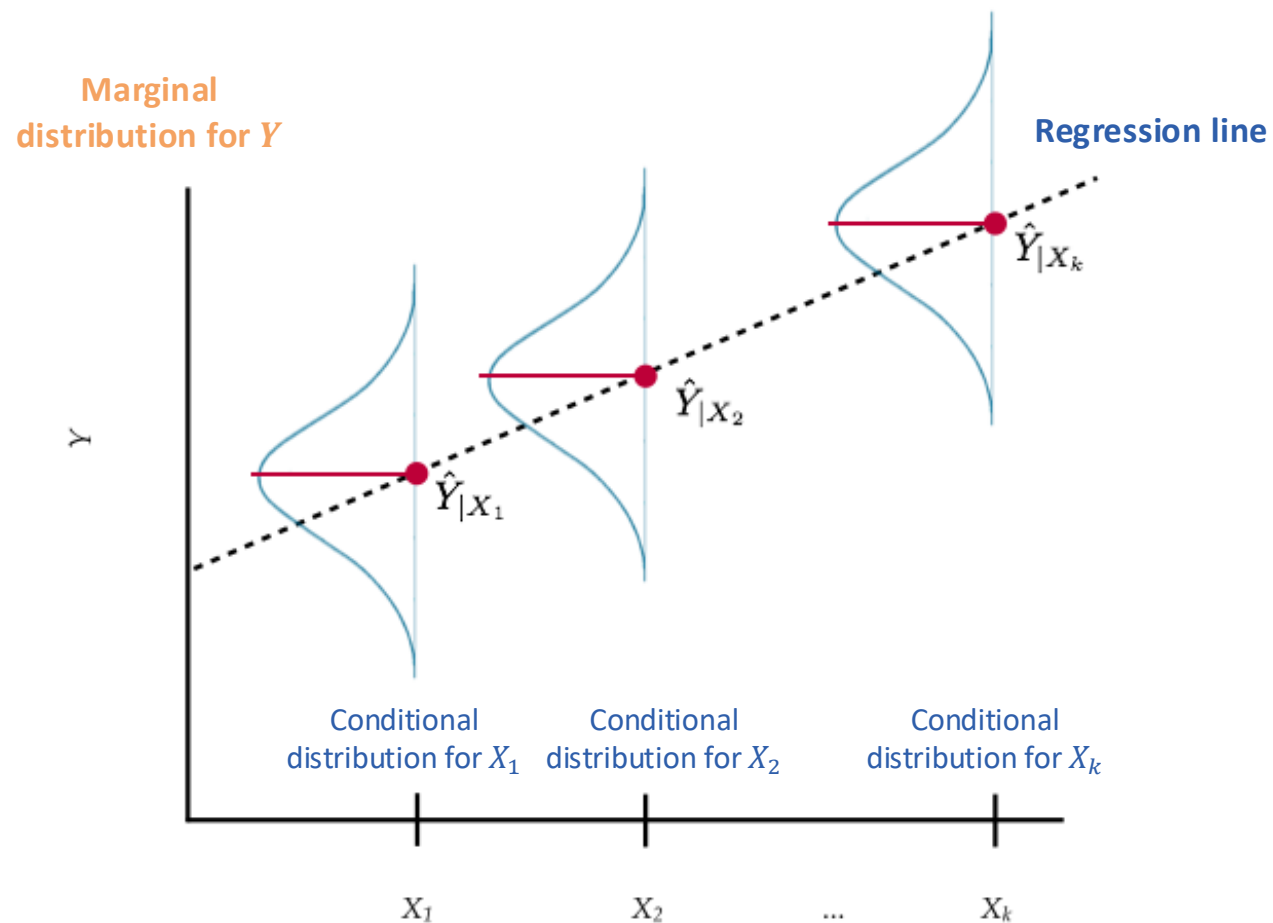
It facilitates **decision-making** and enhances understanding of the problem at hand.

Definition

Based on the **conditional distribution** $Y|X = x_t$ with parameters:

$$E(Y|X = x_t) = \beta_0 + \beta_1 X_t$$

$$\sigma^2(Y|X = x_t) = \sigma^2 \text{ cte}$$



Definition

If we consider the **difference** (u_t) between the value observed (y_t) and the mean of the explanatory variable (x_t).

$$u_t = y_t - (\beta_0 + \beta_1 X_t)$$

Where:

$$E(u_t) = 0 \text{ and } \sigma^2(u_t) = \sigma^2$$

It is assumed that u_t are normally distributed and independent of one another.

The relationship between the residual variance of the model σ^2 and the variance of the marginal distribution of Y will serve as an indicator of the importance of all these factors excluded from the model.

Definition

Finally, the simple linear regression model can be defined as:

$$Y_t = \beta_0 + \beta_1 X_t + u_t$$

Depending on the basic hypothesis, different generalizations can be developed:

- 1) Inclusion of more than one explanatory variable.
- 2) Consideration of non-linear relationships.
- 3) Inclusion of qualitative explanatory variables.

Generalization (1)

Inclusion of more than one explanatory variable:

$$E(Y|X_1 = x_{1t}, X_2 = x_{2t}, \dots, X_I = x_{It}) = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \dots + \beta_I X_{It}$$

Then, the equation for the model would be:

$$Y_t = \beta_0 + \beta_1 X_{1t} + \beta_2 X_{2t} + \dots + \beta_I X_{It} + u_t$$

Where:

- β_0 is the mean value of Y when $X_1 = 0, X_2 = 0, \dots, X_I = 0$
- β_i is increase in the mean value of Y when X_i increases by one unit, with the other explanatory variables held constant.

Generalization (2)

Consideration of non-linear relationships

So far, the analysis has assumed that the possible effect of the explanatory variables on the mean value of Y is linear.

It is interesting to consider more general effects in the model, such as **polynomial functions**, by **introducing these powers** as if they were **new explanatory variables**.

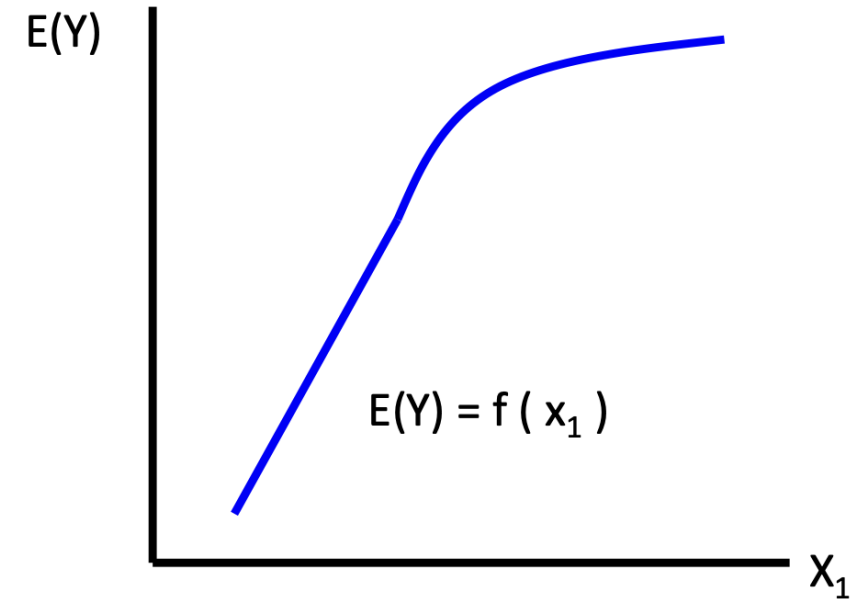
The model $E(Y) = \beta_0 + \beta_1 X_1$ assumes that if X_1 influences Y , it does so linearly. It is **possible that the relationship is non-linear**: progressive increases in X_1 , result in progressively smaller increases in the mean value of Y .

Generalization (2)

Quadratic relationships:

A non-linear relationship can often be approximated by a quadratic curve $a + bX + cX^2$. We define a “new” variable $X_2 = X_1^2$:

$$E(Y) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2$$



Possible non-linear relationship
between E(Y) and X₁

Generalization (1)

Quadratic relationships:

$$Y_t = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + u_t$$

Where:

- β_0 is the mean value of Y when $X_1 = 0$
- β_1 is a slope of the regression line at the origin (roughly equal to the increase in $E(Y)$ when X_1 changes from 0 to 1, if the range of variation of X_1 is several units).
- β_2 is a measure of the curvature (positive or negative) of the relationship between $E(Y)$ and X_1 . Thus, a **positive value of β_2** would indicate that the slope of the curve relating $E(Y)$ to X_1 increases as X_1 increases (**positive curvature**), whereas if β_2 were negative, that slope would decrease as X_1 increases (**negative curvature**).

Generalization (2)

Interaction of linear effects:

$$Y_t = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 \cdot X_2) + u_t$$

Where:

- β_0 is the mean value of Y when $X_1 = 0$ and $X_2 = 0$.
- β_1 is the increase in $E(Y)$ when X_1 changes from 0 to 1.
- β_2 is the increase in $E(Y)$ when X_2 changes from 0 to 1.
- β_3 is the interaction between the linear effects of X_1 and X_2 . This indicates by how much the linear effect of one variable increases for every unit increase in the other.

Generalization (3)

Inclusion of qualitative explanatory variables:

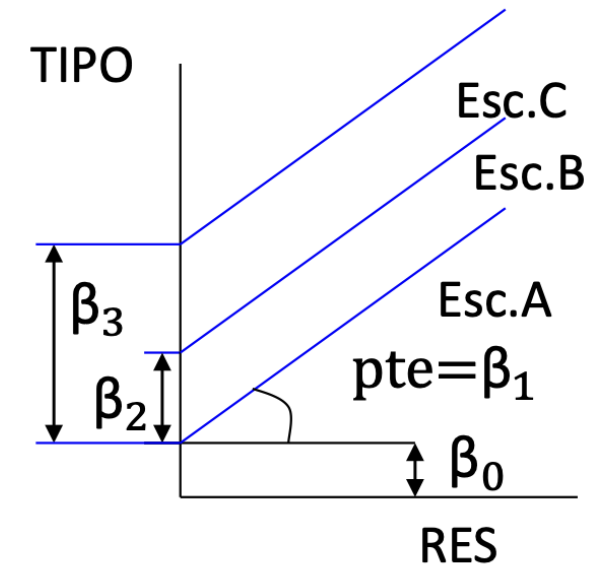
You should **never treat a qualitative variable** with more than two possible values **as if it were quantitative**.

The correct approach is to include as many dummy variables (with possible values of 0 or 1) as there are categories in the qualitative variable, minus one. With K categories, **$(K - 1)$ dummy variables** would be included.

Generalization (3)

Having a categorical variable X with 3 categories ($K=3$) we would create 2 dummy variables (Z_1 and Z_2)

Categories	Z_1	Z_2
A	0	0
B	1	0
C	0	1



Generalization (3)

Inclusion of qualitative explanatory variables:

$$Y_t = \beta_0 + \beta_1 X_1 + \beta_2 Z_1 + \beta_3 Z_2 + u_t$$

$$E(Y_t | X_1, A) = \beta_0 + \beta_1 X_1$$

$$E(Y_t | X_1, B) = \beta_0 + \beta_1 X_1 + \beta_2$$

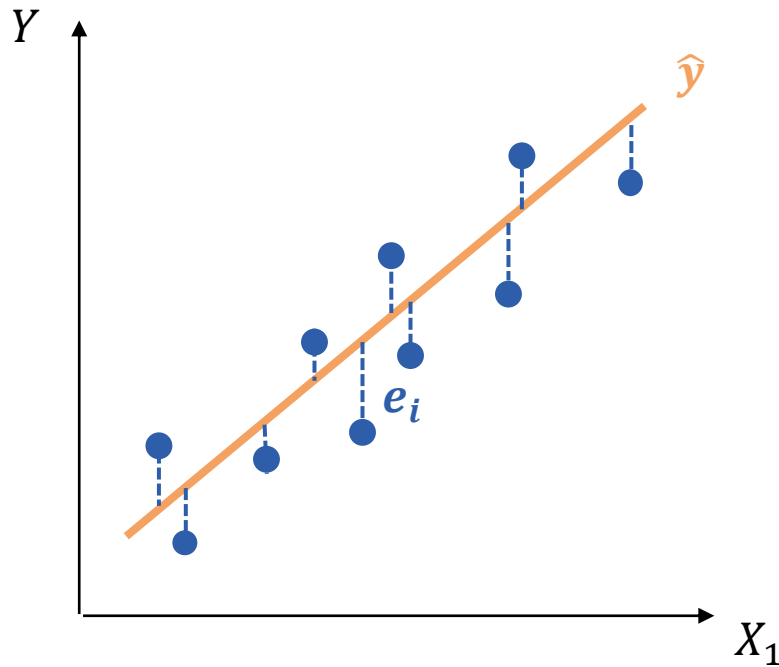
$$E(Y_t | X_1, C) = \beta_0 + \beta_1 X_1 + \beta_3$$

Where:

- β_2 is the average category difference obtained **using B** compared to when using A (for the same resolution X_1).
- β_3 is the average category difference obtained **using C** compared to when using A (for the same resolution X_1).

Inference in regression models

It is based on the estimation of the coefficients β_i and their residual variances s_{β_i} . The estimation results provide the corresponding linear regression equation ($\hat{y}$).



Parameters estimation:

$$\beta_0 = r \cdot \frac{s_Y}{s_X}$$
$$\beta_1 = \bar{y} - \beta_0 \bar{X}_1$$

Minimization of the error:

$$\min \sum_{i=1}^N (y_i - (\beta_0 + \beta_1 X_1))^2 = \min \sum_{i=1}^N e_i^2$$

Regression line:

$$\hat{y} = \beta_0 + \beta_1 X_1$$

Overall significance of the fit.

Total variability of the dependent variable Y across all N observations :

$$SCT = \sum_i^N (y_i - \bar{y})^2$$

Where $\bar{y}$ is the mean of dependent variable Y , and $gl_{total} = I$.

Overall significance of the fit.

Part of the variability in Y **associated with the residuals**:

$$SCR = \sum_i^N e_i^2$$

Where e_j^2 is the error, and $gl_{residual} = (N - 1) - I$.

The Residual Mean Square (CMR) is an estimate of the model's residual variance:

$$\sigma^2 \approx CMR = \frac{SCR}{gl}$$

Overall significance of the fit.

Part of the variability in Y **associated with the explanatory variables:**

$$SC_{EXP} = SCT - SCR$$

Overall significance of the fit.

Hypothesis testing:

$$\begin{cases} H_0: \beta_1 = \beta_2 = \dots = \beta_I = 0 \rightarrow \text{Accepted when } \frac{CM_{EXP}}{CMR} < F_{I, N-1-I} \\ H_1: \exists \beta_i \neq 0 \end{cases}$$

If the null hypothesis is accepted:

$$\frac{SC_{EXP}/I}{SCR/(N-1-I)} = \frac{CM_{EXP}}{CMR} \sim F_{I, N-1-I}$$

R² coefficient

$$R^2 = \frac{SC_{model}}{SC_{total}} = 1 - \frac{CM_{residual}}{CM_{total}} \quad \text{where } R^2 \in [0, 1]$$

The **closer the value is to 1**, the greater the proportion of the observed variability in (*Y*) that is explained by the explanatory variables (*X_i*) included in the model.

Significance of the effect of a variable

Hypothesis testing:

$$\begin{cases} H_0: \beta_i = 0 \rightarrow \text{Accepted when } \frac{b_1}{s_{b_i}} < t_{N-1-I} \\ H_1: \beta_i \neq 0 \end{cases}$$

If the null hypothesis is accepted ($\beta_i = 0$):

$$\frac{b_1}{s_{b_i}} \sim t_{N-1-I}$$

Where b_1 is the sample estimate of the parameter β_i .

Residual analysis

$$Y_t = \beta_0 + \beta_1 X_{1t} + \cdots + \beta_I X_{It} + u_t$$

Where the residuals are independent and follow a normal distribution:

$$u_t \sim N(0, \sigma^2)$$

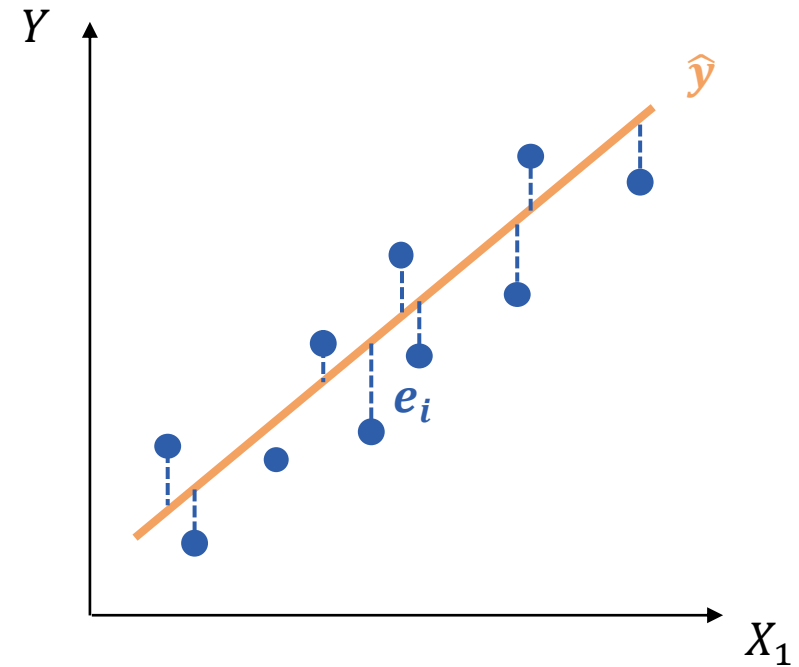
- Is it reasonable to assume that the **Y values are normally distributed?**
- Are there any clearly **anomalous data** points?
- Is it reasonable to assume that the variance of the Y values does not depend on the values of the explanatory variables?
- Is the **relationship between $E(Y)$ and an X_i truly linear?**

Residual analysis

For answering all these question a residual analysis should be done:

$$e_t = y_0 - (\beta_0 + \beta_1 X_{1t} + \dots + \beta_I X_{It})$$

The residuals e_t are estimates of the values of the random disturbances u_t at each observation.



Applications:

1. A plot of the e_t values on standard probability paper allows us to assess whether the assumption of normality holds, as well as to identify any outliers.
2. A plot of the squared residuals against the various X_i variables may reveal that some of the explanatory variables influence the variance of the Y_t values.
3. A plot of the e_t values against the predicted values for each observation $\beta_0 + \beta_1 X_{1t} + \dots + \beta_I X_{It} + u_t$ can reveal the existence of non-linear relationships.

Case Study 1

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 Z_1 + \beta_4 Z_2 + \beta_5 X_1 X_2 + u_t$$

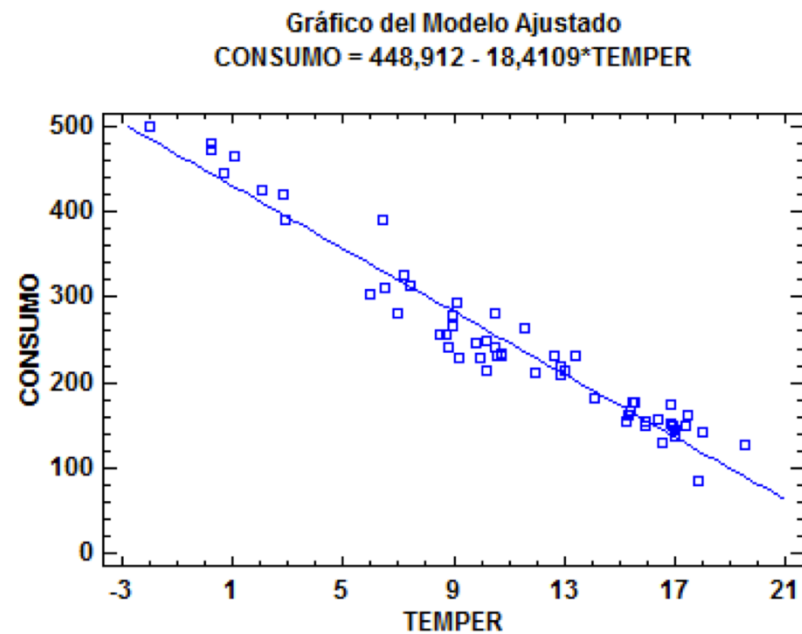
Where:

- Y is the daily energy consumption in winter at a factory.
- X_1 is the temperature on the day in question
- X_2 is the number of employees working on the day in question
- X_3 is the type of shift:
 - A. Morning shift
 - B. Afternoon shift
 - C. Night shift

Categories	Z_1	Z_2
A	1	0
B	0	0
C	0	1

Case Study 2

Linear Model



Coeficientes

	<i>Mínimos Cuadrados</i>	<i>Estándar</i>	<i>Estadístico</i>	
<i>Parámetro</i>	<i>Estimado</i>	<i>Error</i>	<i>T</i>	<i>Valor-P</i>
Intercepto	448,912	7,63267	58,8145	0,0000
Pendiente	-18,4109	0,627143	-29,3567	0,0000

Análisis de Varianza

<i>Fuente</i>	<i>Suma de Cuadrados</i>	<i>Gl</i>	<i>Cuadrado Medio</i>	<i>Razón-F</i>	<i>Valor-P</i>
Modelo	552051,	1	552051,	861,82	0,0000
Residuo	35231,1	55	640,566		
Total (Corr.)	587282,	56			

Coefficiente de Correlación = -0,969541

R-cuadrada = 94,001 por ciento

R-cuadrado (ajustado para g.l.) = 93,8919 por ciento

Error estándar del est. = 25,3094

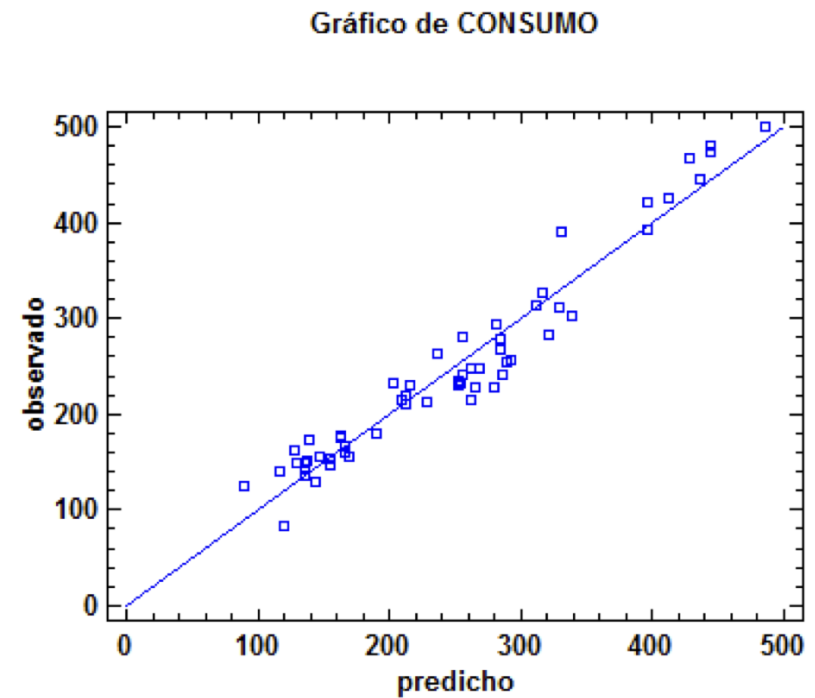
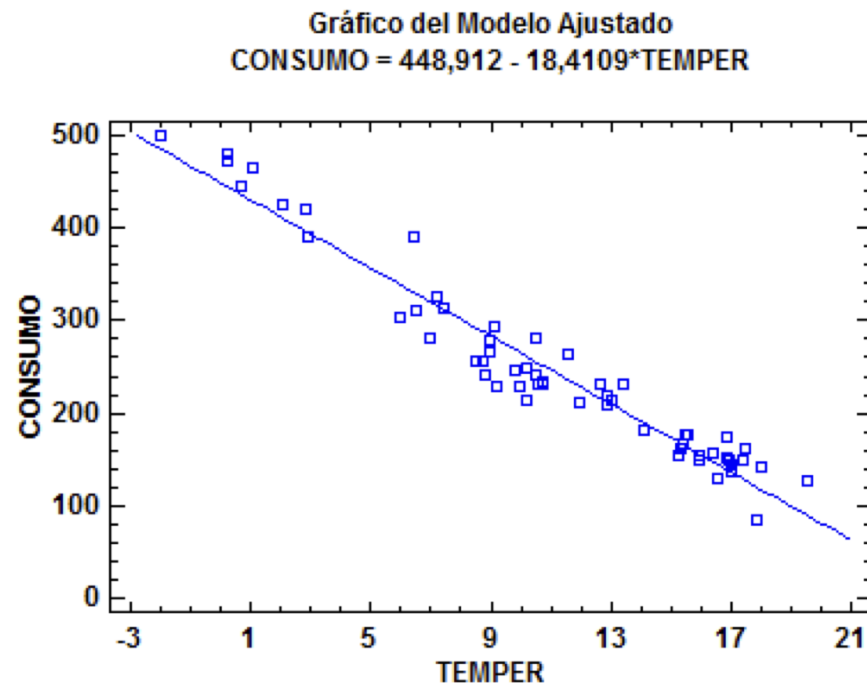
Error absoluto medio = 20,5716

Estadístico Durbin-Watson = 1,38718 (P=0,0062)

Autocorrelación de residuos en retraso 1 = 0,292713

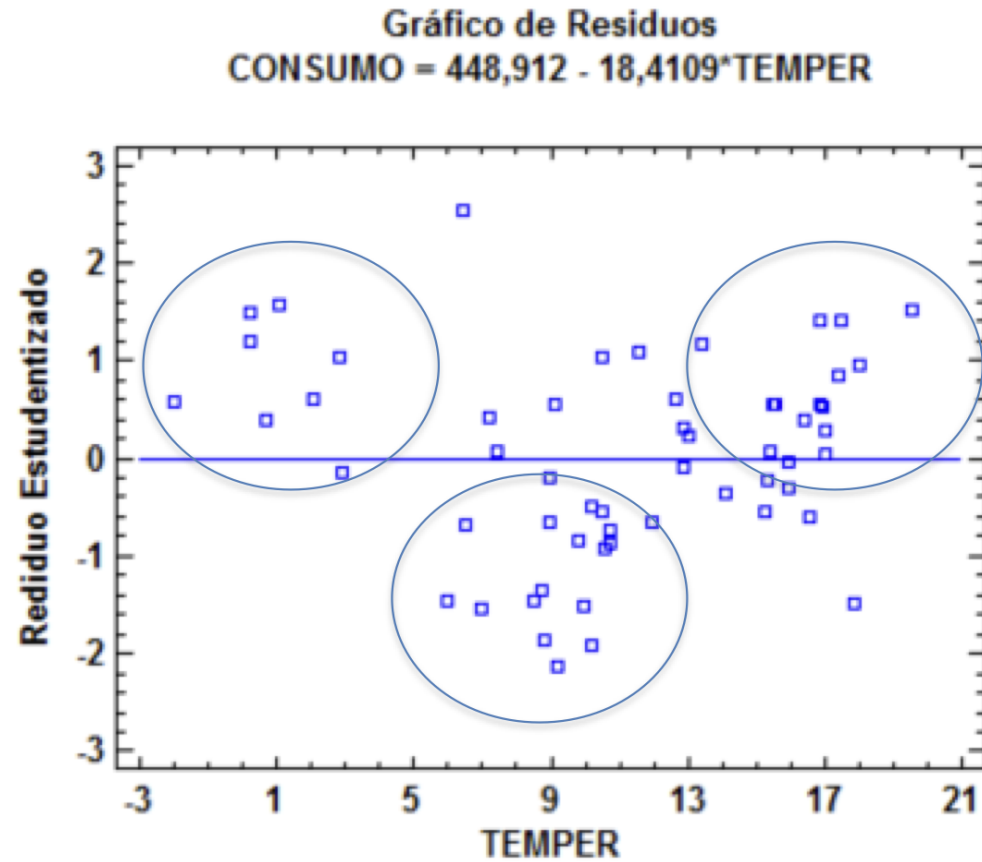
Case Study 2

Linear Model



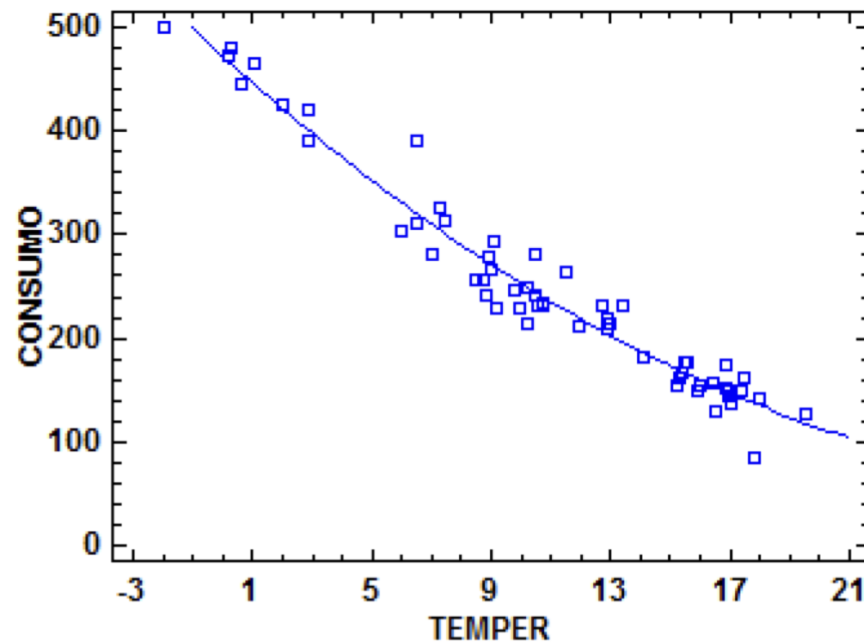
Case Study 2

Linear Model



Quadratic Model

Gráfica del Modelo Ajustado



		Error	Estadístico	
Parámetro	Estimado	Estándar	T	Valor-P
CONSTANTE	472,35	8,56846	55,1266	0,0000
TEMPER	-25,9864	1,83412	-14,1683	0,0000
TEMPER^2	0,400966	0,092686	4,32607	0,0001

Análisis de Varianza

Fuente	Suma de Cuadrados	Gl	Cuadrado Medio	Razón-F	Valor-P
Modelo	561118,	2	280559,	579,06	0,0000
Residual	26163,6	54	484,51		
Total (Corr.)	587282,	56			

R-cuadrada = 95,545 por ciento

R-cuadrada (ajustada por g.l.) = 95,38 por ciento

Error estándar del est. = 22,0116

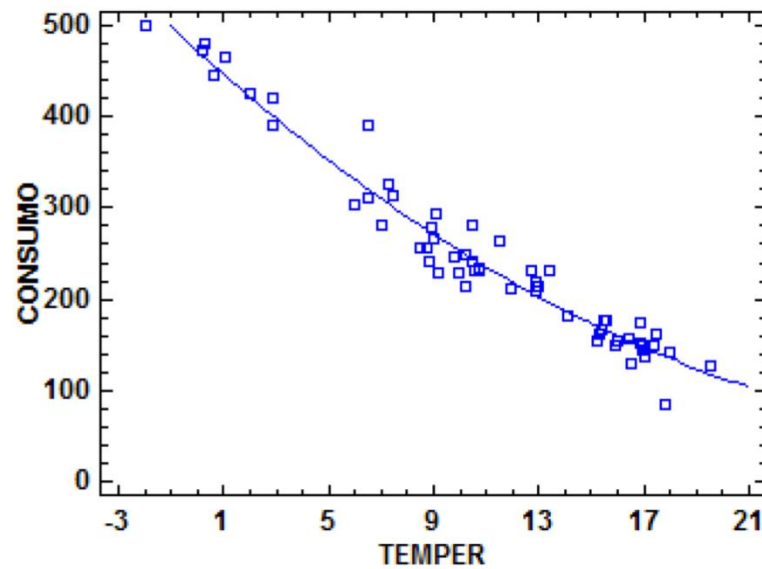
Error absoluto medio = 16,5403

Estadístico Durbin-Watson = 1,52383 (P=0,0205)

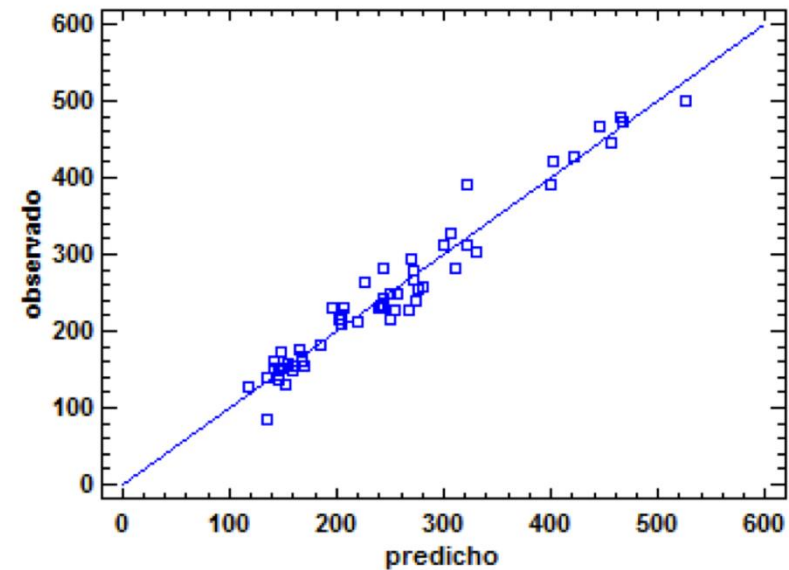
Autocorrelación de residuos lag 1 = 0,200748

Quadratic Model

Gráfica del Modelo Ajustado

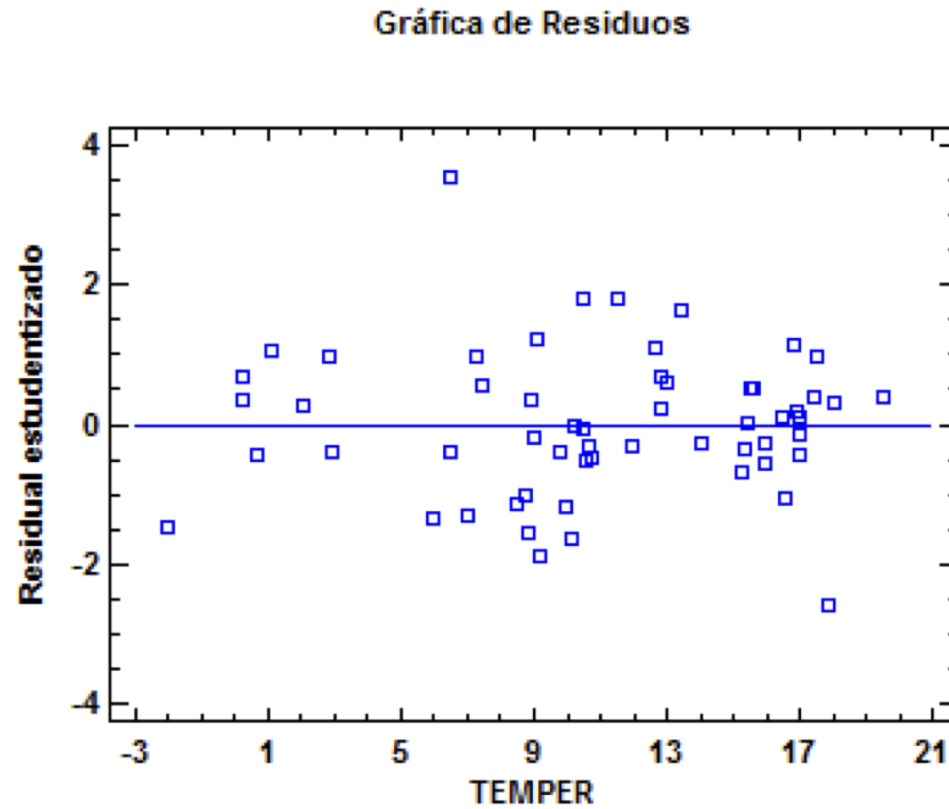


Gráfica de CONSUMO



Case Study 2

Quadratic Model



Exercise 1

In the production of a jam, the aim is to investigate how the content of a specific nutritional microcomponent (mg) responds to the quantity of one of the raw materials (MAT1) used in its manufacture, which can be supplied via three different formulations (A, B, C) that may influence the final amount of the nutritional microcomponent incorporated. To carry out the study, the content of the micro-nutrient was measured in 10 jars produced using each of the three formulations (30 jars in total) across different runs of the process, in which varying quantities of raw material MAT1 were used, selected from the range of 0 g to 100 g. Using the data obtained, the best regression model that could be derived is as follows:

$$Y = \beta_0 + \beta_1 FOR_B + \beta_2 FOR_C + \beta_3 MAT_1 + \beta_4 MAT_1 \cdot FOR_B$$

Exercise 1

Where FOR_B and FOR_C are two dummy variables for the qualitative factor “Formulated”, which take the following values:

FORMULADOS	FORB	FORC
A	0	0
B	1	0
C	0	1

Exercise 1

1) Indicate, using a number from 0 to 4, the parameter that has been misinterpreted. If there is more than one, enter 5. (In the assignment, state what the correct interpretation of the misinterpreted term(s) would be:

1. β_0 : Average microcomponent content when the amount of MAT1 used is 0 and formulation A is employed
2. β_1 : Change in the average microcomponent content when switching from formulation A to formulation B when the amount of MAT1 used is 0
3. β_2 : Change in the average microcomponent content when switching from formulation A to formulation C when the amount of MAT1 used is 0
4. β_3 : Slope of the relationship between average microcomponent content and the amount of MAT1 used for formulations A and C
5. β_4 : Change in the slope of the relationship between the average microcomponent content and the amount of MAT1 used when switching from formulation A to formulations B or C

Exercise 1

Análisis de Regresión Múltiple

Variable depend.: Contenido Microcomponente

Parámetro	Estimación	Error estándar	Estadístico T	P-Valor
CONSTANTE	19,5406	2,37464	8,22887	0,0000
FOR=B	-8,25417	3,76336	-2,1933	0,0382
FOR=C	5,059	1,94339	2,60318	0,0156
MAT1	0,232419	0,0341807	6,79973	0,0000
MAT1*FOR=B	0,467167	0,0585955	7,97274	0,0000

Análisis de Varianza					
Fuente	Suma de cuadrados	GL	Cuadrado medio	Cociente-F	P-Valor
Modelo	10375.087	4	2593.77	143,07	0,0000
Residuo	453.213	25	18,1285		
Total (Corr.)	10828,3	29			

Exercise 1

- 2) Obtain a graph showing the relationship between the average micro-nutrient content and the amount of MAT1.
- 3) From what amount of MAT1 does Formulation B result in a higher micro-nutrient content than the others?
- 4) What percentage of the variability in micro-nutrient content would be explained by the proposed model?
- 5) A company technician takes a container to which 25 g of MAT1 has been added during production using formulation C and observes that the final content of the incorporated nutrient microcomponent is 24 mg. What is the residual?

Exercise 2

In a reaction for the synthesis of a certain product, the reaction rate appears to depend on the temperature and the catalyst used. To investigate the nature of this relationship, a sample of 80 runs of the process was obtained and a multiple linear regression model was estimated, where the response variable is the 'reaction rate' measured in units of concentration per second (u.c/s) and the explanatory variables are:

- **Temperature:** the constant at which the process is maintained during the reaction. In the sample, the 80 runs were carried out at different temperatures within the range from 0°C to 120°C, given that the reaction does not take place below 0°C and undesirable processes would occur at temperatures above 120°C.
- **Catalyst:** there are two possible catalysts on the market (A and B).

Exercise 2

The first model estimated in the analysis was as follows:

$$Y = \beta_0 + \beta_1 TEMP + \beta_2 TEMP^2 + \beta_3 CAT_A + \beta_4 CAT_A \cdot TEMP + \beta_5 CAT_A \cdot TEMP^2$$

Where:

- **β_0** : Average reaction rate when the temperature is 0 °C and Catalyst B is used
- **β_2** : Curvature of the relationship between the average reaction rate and temperature when Catalyst B is used
- **β_4** : Difference in the slope of the relationship between the average reaction rate and temperature when the temperature is 0 °C and Catalyst A is used, compared to when Catalyst B is used
- **β_5** : Difference in the curvature of the relationship between the average reaction rate and temperature when Catalyst A is used, compared to when Catalyst B is used

Exercise 2

The following Statgraphics output shows the final estimated model, retaining only those terms whose parameters have been found to be significantly different from zero:

Variable dependiente: $\bar{\text{Veloc Reaccion}}$

Variables independientes: T ; $\text{CAT}_A * T$; $\text{CAT}_A * T^2$

		Error	Estadístico	
Parámetro	Estimación	Estándar	T	Valor-P
CONSTANTE	5,91761	0,562778	10,515	0,0000
T	0,332205	0,00780182	42,5804	0,0000
$\text{CAT}_A * T$	-0,368298	0,0222212	-16,5742	0,0000
$\text{CAT}_A * T^2$	0,005562	0,000239414	23,2318	0,0000

Análisis de Varianza

Fuente	Suma de Cuadrados	Gl	Cuadrado Medio	Razón-F	Valor-P
Modelo	24079,2				
Residuo					
Total (Corr.)	24373,3				

Exercise 2

- a) Interpret the parameters β_1 and β_3 of the proposed model.
- b) Find the value of the coefficient of determination and explain its significance.
- c) Plot an approximate graph of the relationship between the average reaction rate and temperature for the two catalysts.
- d) It is known that above 90°C the synthesis reaction can lead to the formation of contaminating by-products. If the aim is to maximise the reaction rate without producing contaminants: at what temperature should the synthesis be carried out, which catalyst should be used, and what would be the probability that the obtained reaction rate does not exceed 50 u.c/s?

Exercise 2

e) Indicate whether the following statements are correct or incorrect and explain why: (answer in a concise and reasoned manner, supporting your reasoning with calculations where necessary)

When examining a data point with a residual of -4 u.c/sec, three technicians have differing opinions. The first states that the data point is an outlier because it has a negative residual. The second technician states that it is an outlier because a residual with an absolute value of 4 seems large. The third disagrees with the others and states that simple calculations show it to be a perfectly reasonable data point.

Exercise 3

To analyse the effect of the type of camera (X_1) and image resolution (X_2) on image processing time (Y), a 2×2 factorial experiment was conducted with three treatments:

- Variants of factor X_1 , scanner type: A and B.
- Levels of factor X_2 , image resolution: 150 and 300 dpi (dots per inch)

a) Formulate a linear regression model to examine the effect of scanner type, the linear effect of image resolution, and their interaction on average processing time. Write out the theoretical model and the estimated model, plotting them and interpreting the meaning of each of their terms and coefficients. Use scanner type A as a reference

Exercise 3

- b) What would the R^2 of the fitted regression model indicate?
- c) What would be the critical value in the tables for the corresponding statistic to test the statistical significance of the linear effect of image resolution at a Type I risk of 1%? Justify your answer.
- d) To test the normality hypothesis, would it be appropriate to plot the processing times obtained in the experiment on a normal probability paper (NPP)? Justify your answer.
- e) Assuming that the variability in processing time increases with image resolution, sketch what the plot of residuals versus image resolution might look like.
- f) What type of analysis should we then carry out to test the hypothesis of homoscedasticity?

Exercise 4

A linear regression analysis was carried out to investigate pharmaceutical expenditure (variable GASTO, in € per week) in relation to the patient's gender (variable SEXO, equal to 0 for men and 1 for women), the patient's age (variable TIEMPO in years) and their monthly salary (variable SUELDO in €).

The results obtained using STATGRAPHICS for the fitting of different models are shown in the next slide (incomplete results):

Exercise 4

MODELO 1

Variable dependiente: GASTO

Parámetro	Estimación	Error estándar	Estadístico T	P-Valor
CONSTANT	0.346167	0.053202	6.5066	0.0000
TIEMPO	-0.027194	0.00724	-3.7562	0.0021
SEXO	0.168	0.043439	3.8675	0.0017
SUELDO	0.054167	0.043439	1.2470	0.2329

SE= 0.0657 MAE= 0.054370 DurWat= 2.049

Análisis de Varianza

Fuente	Suma de cuadrados	GL	Cuadrado medio	Cociente-F	P-Valor
Modelo	_____	_____	_____	_____	0.000
Error	_____	_____	_____		

Total (Corr.) 0.247355 17

R-cuadrado = 0.679599

R-cuadrado (ajustado para g.l.) = 0.70103

Exercise 4

MODELO 2

Variable dependiente: GASTO

Parámetro	Estimación	Error estándar	Estadístico T	P-Valor
CONSTANTE	0.210083	0.018986	11.0652	0.0000
TIEMPO	0.0535	0.010962	4.8807	0.0002
SEXO	0.461917	0.073532	6.2818	0.0000

SE=	0.065769	MAE=	0.046676	DurbWat= 1.621

Análisis de Varianza

Fuente	Suma de cuadrados	GL	Cuadrado medio	Cociente-F	P-Valor
Modelo	0.182471	2	0.0912355	21.0920	0.0000
Error	0.0648839	15	0.00432559		
Total (Corr.)	0.247355	17			
R-cuadrado = 0.737689			Error estándar de est. = 0.0657693		
R-cuadrado (Adj. for d.f.) = 0.702714					

Exercise 4

- a) What is the equation for Model 1?
- b) What is the coefficient of determination for Model 1?
- c) Is Model 1 fully significant? Justify your answer.
- d) Is Model 2 fully significant? Justify your answer.
- e) Which of the two models would be justified to choose from a statistical point of view?
- f) Assuming Model 1 is chosen, estimate the expenditure of male patients with a monthly salary of €1,000 and aged 55.

Exercise 5

In the production of a jam, the aim is to investigate how the content of a specific nutritional microcomponent (mg) responds to the quantity of one of the raw materials (MAT1) used in its manufacture, which can be supplied via three different formulations (A, B, C) that may influence the final amount of the nutritional microcomponent incorporated. To carry out the study, the content of the micro-nutrient was measured in 10 jars produced using each of the three formulations (30 jars in total) across different runs of the process, in which varying quantities of raw material MAT1 were used, selected from the range of 0 g to 100 g. Based on the data obtained, the best regression model that could be derived is as follows:

$$Y = \beta_0 + \beta_1 FOR_B + \beta_2 FOR_C + \beta_3 MAT_1 \cdot FOR_B + \beta_4 MAT_1 \cdot FOR_C$$

Exercise 5

Where FOR_B and FOR_C are two dummy variables for the qualitative factor “Formulated”, which take the following values:

FORMULADOS	FORB	FORC
A	0	0
B	1	0
C	0	1

Exercise 5

Análisis de Regresión Múltiple

Variable depend.: Contenido Microcomponente

Parámetro	Estimación	Error estándar	Estadístico T	P-Valor
CONSTANTE	19,5406	2,37464	8,22887	0,0000
FOR=B	-8,25417	3,76336	-2,1933	0,0382
FOR=C	5,059	1,94339	2,60318	0,0156
MAT1*FOR=B	0,467167	0,0585955	7,97274	0,0000
MAT1*FOR=C	-0.23242			0,0000

Análisis de Varianza

Fuente	Suma de cuadrados	GL	Cuadrado medio	Cociente-F	P-Valor
Modelo			2068.77	93.49	0,0000
Residuo	553,213		22.1285		
Total (Corr.)	8828,3	29			

Exercise 5

- a) Obtain a graph showing the relationship between the average micro-nutrient content and the amount of MAT1
- b) At what level of MAT1 does Formulation B result in a higher nutrient content than the others?
- c) What percentage of the variability in micro-nutrient content would be explained by the proposed model?